Worksheet 2
Assembling Plots

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After gaining an understanding of the grammar of graphics in the last worksheet, in this one we want to understand how the concept is implemented in ggplot2. ggplot2 defines a lot of sensible defaults and shorthands that are in practice frequently used and can make the code a lot more concise and elegant.

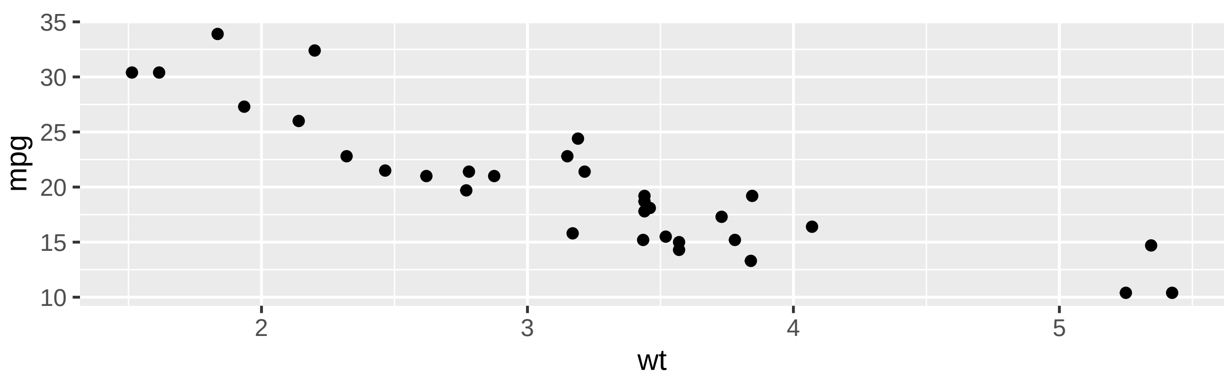
If you are looking for information about which default is associated with any grammar or stats, or about any other ggplot2 function, the documentation at ggplot2.tidyverse.org is a very helpful resource.

1 Understanding and modifying plots

Grab a partner, and take take a look at some of the plots below. You've already seen these in Worksheet 1. Remember what you observed earlier, and how this is reflected in the code that produces these plots. Below, there are some suggestions on how to modify these plots. If you don't have R installed, you can try this out online at rdrr.io/snippets/. Remember to import ggplot with library(ggplot2) before plotting! Feel free to also try other modifications or plot types and play around with the data!

1.1 Plot 1

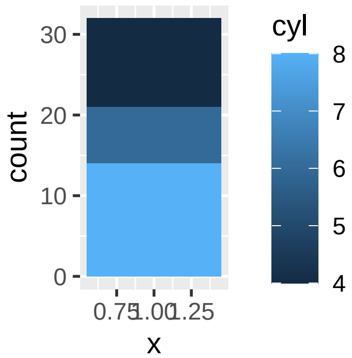
```
ggplot(mtcars) +
    geom_point(aes(x=wt, y=mpg))
```



- Plot the horsepower (hp) instead of the fuel efficiency! - Try setting the colours or shapes of the points to how many cylinders each car has – which works best? - Can you convert the plot to use litres per 100 km instead of miles per gallon? - Which other geometries could be suitable to display this data? Try some out!

1.2 Plot 2

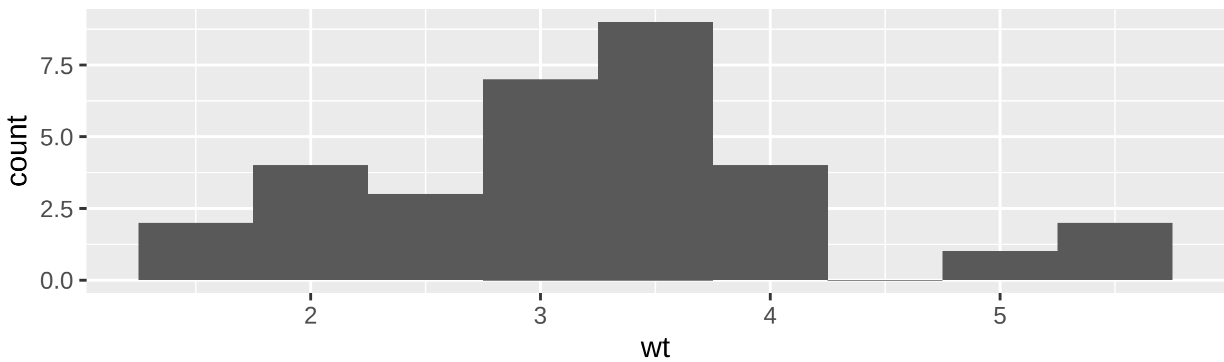
```
ggplot(mtcars) +
    geom_bar(aes(x=1, group=cyl, fill=cyl))
```



- What happens if you set the colour instead of the fill?
- The exact plot in Worksheet 1 looks a little bit different. What is the difference, and how can you make it look the same?
- These bars are all stacked. Can you find a way to plot them next to each other?
- Can you make a pie chart out of this plot?

1.3 Plot 3

```
ggplot(mtcars) +
    geom_histogram(aes(wt), binwidth=0.5)
```

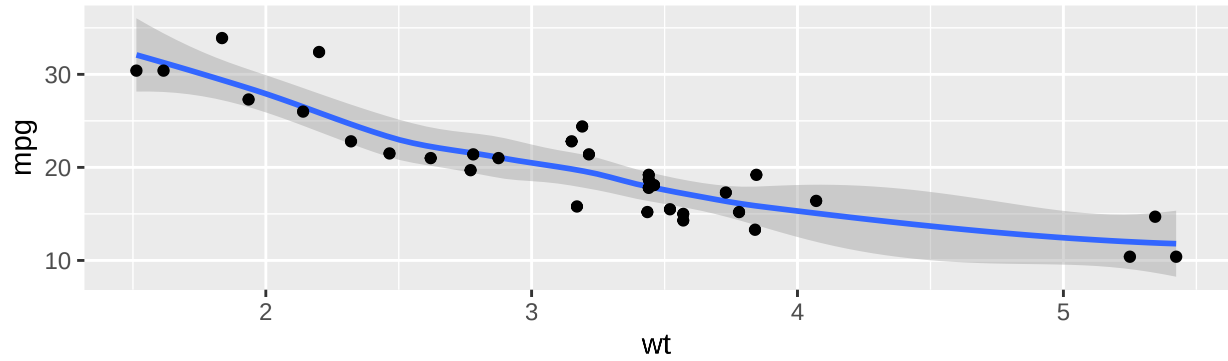


- What happens as you change the binwidth or amount of bins?
- Can you display the distribution of other features in this plot as well? Try the amount of cylinders!
- Compare this plot to what you get with `geom_density`! What is the advantage of either?

1.4 Plot 4

```
ggplot(mtcars) +
    geom_smooth(aes(x=wt, y=mpg)) +
    geom_point(aes(x=wt, y=mpg))
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



- There seems to be an association between the two plotted variables. How can you investigate it using ggplot?
- Do other variables in the dataset affect this association? Try adding them into the plot and have a look!

2 Designing new plots

Grab a partner, and think about how to answer the questions given below by cleverly using one or multiple plots. Think about a visualization first, write it down and then try implementing it using ggplot. You can again use `rdrr.io/snippets/` if you do not have R installed. Remember to import ggplot with `library(ggplot2)` before plotting!

2.1 Which car brand is the best?

Which brand makes the best cars? Try to compare brands using different criteria, and assemble plots illustrating why a given brand is best or not ideal for each. For extracting the car's brand from its name, you can use the code given below:

```
library(stringr)
mtcars$brand <- word(rownames(mtcars), 1)
```

2.2 What makes a diamond's price?

Try to find out why a diamond is worth its price! Start by investigating the relationship between price and carat. Do other factors play a role? When plotting, think of how to best handle big datasets with a large spread in values! ngk"k"